

EduVision Programming

New Student Assessment Paper

Basic Information

Name: Aaron Sun

Current Year: 8

Contact Number: _____

Past experience learning C++ (terms): _____

Past experience learning Python (terms): _____

Part one --- Self-Assessment Trial Questions

These questions are optional. You are encouraged to attempt them. Please submit the source code together with the completed questionnaire.

Problem 1: Numbers Containing Digit 3 or 7

Given two integers m , n , find the number of integers from m to n (inclusive) which contain the digit 3 or 7.

Input Format

One line containing two integers m , n separated by spaces.

Output Format

Output one line containing one integer which is the number of integers from m to n that contain the digit 3 to 7.

Example

Input #1:

13 37

Output #1:

9999999999

Output #2:

12656242944

Constraints

$1 \leq |s| \leq 10$

Part two ----- Circle your answer next to each question

Syntax and Standard Library

Input, Output, and Variables:	Not learned	Learned	Proficient
Conditional Branching:	Not learned	Learned	Proficient
Loops:	Not learned	Learned	Proficient
One-Dimensional Arrays:	Not learned	Learned	Proficient
Two-Dimensional Arrays:	Not learned	Learned	Proficient
Character Arrays:	Not learned	Learned	Proficient
Strings:	Not learned	Learned	Proficient
Custom Functions:	Not learned	Learned	Proficient
Struct:	Not learned	Learned	Proficient
Files and Basic I/O:	Not learned	Learned	Proficient
STL Templates:	Not learned	Learned	Proficient

Algorithms and Data Structures (Entry Level)

Enumeration:	Not learned	Learned	Proficient
Simulation:	Not learned	Learned	Proficient
Bubble Sort:	Not learned	Learned	Proficient
Quick Sort:	Not learned	Learned	Proficient
Queues:	Not learned	Learned	Proficient

Stacks:	Not learned	Learned	Proficient
Linked Lists:	Not learned	Learned	Proficient
Trees and Binary Trees:	Not learned	Learned	Proficient
Binary Search:	Not learned	Learned	Proficient
Binary Lifting	Not learned	Learned	Proficient
High-Precision Arithmetic	Not learned	Learned	Proficient
Recursion	Not learned	Learned	Proficient
Greedy Algorithms	Not learned	Learned	Proficient
Depth-First Search (DFS)	Not learned	Learned	Proficient
Breadth-First Search (BFS)	Not learned	Learned	Proficient
Basics of Dynamic Programming (DP)	Not learned	Learned	Proficient
1D Dynamic Programming	Not learned	Learned	Proficient
Knapsack Problems	Not learned	Learned	Proficient
Interval DP	Not learned	Learned	Proficient
Euclidean Algorithm	Not learned	Learned	Proficient
Prime Sieves: Sieve of Eratosthenes			
& Linear Sieve	Not learned	Learned	Proficient
Sets	Not learned	Learned	Proficient
Addition Principle (Rule of Sum) &			
Multiplication Principle (Rule of Multiplication)	Not learned	Learned	Proficient
Permutations and Combinations	Not learned	Learned	Proficient
STL	Not learned	Learned	Proficient
Prefix Sum	Not learned	Learned	Proficient
Difference Array	Not learned	Learned	Proficient

Algorithms and Data Structures (Advanced Level)

Shortest Path	Not learned Learned Proficient
Single-Source Second Shortest Path	Not learned Learned Proficient
Minimum Spanning Tree (MST)	Not learned Learned Proficient
Second Minimum Spanning Tree	Not learned Learned Proficient
Topological Sorting	Not learned Learned Proficient
Eulerian Path and Eulerian Circuit	Not learned Learned Proficient
Bipartite Graph Determination	Not learned Learned Proficient
Strongly Connected Components (SCC)	Not learned Learned Proficient
Articulation Points & Bridges	Not learned Learned Proficient
Union-Find (Disjoint Set Union)	Not learned Learned Proficient
Fenwick Tree (Binary Indexed Tree)	Not learned Learned Proficient
Segment Tree	Not learned Learned Proficient
Splay Tree	Not learned Learned Proficient
Treap	Not learned Learned Proficient
Center, Diameter and Centroid of a Tree	Not learned Learned Proficient
DFS Order and Euler Tour of a Tree	Not learned Learned Proficient
Tree Difference, Subtree Sum & Doubling	Not learned Learned Proficient
Lowest Common Ancestor (LCA)	Not learned Learned Proficient
Discretisation	Not learned Learned Proficient
Divide and Conquer	Not learned Learned Proficient
Radix Sort	Not learned Learned Proficient
Tree DP	Not learned Learned Proficient
State Compression DP	Not learned Learned Proficient
Common DP Optimisations	Not learned Learned Proficient

Euler's Theorem & Euler's Totient Function	Not learned Learned Proficient
Bézout's Identity	Not learned Learned Proficient
Modular Multiplicative Inverse	Not learned Learned Proficient
Extended Euclidean Algorithm	Not learned Learned Proficient
Chinese Remainder Theorem	Not learned Learned Proficient
Permutations & Combinations on Multisets	Not learned Learned Proficient
Derangements & Circular Permutations	Not learned Learned Proficient
Inclusion-Exclusion Principle	Not learned Learned Proficient
Catalan Numbers	Not learned Learned Proficient
Vectors and Matrices	Not learned Learned Proficient
Gaussian Elimination	Not learned Learned Proficient
String Matching (KMP)	Not learned Learned Proficient

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